

Neel Maddu

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SUMMARY

I've built AI-powered tools like a multimodal search engine using CLIP, FAISS, and PyTorch, and led a team of 8 to ship production-level iOS apps with Core ML at Georgia Tech. Incoming Software Engineer Intern at GTRI's CIPHER Lab, where I'll apply my ML and systems experience to cybersecurity and intelligence analysis.

EDUCATION

Georgia Institute of Technology

Bachelor of Science in Computer Science | **GPA: 4.0**

Atlanta, GA

Expected May 2027

TECHNICAL SKILLS

Machine Learning: PyTorch, TensorFlow, scikit-learn, FAISS

Data: pandas, NumPy

iOS Development: Swift, SwiftUI, Core ML, MVVM

Backend & Databases: SQL, MySQL, MongoDB, Firebase, Node.js

Programming Languages: Python, Java, C, JavaScript, C++, Next.js, React

Tools: Git, GitHub, CI/CD, Jenkins, GitHub Actions, Docker, CMake

Cloud: AWS, Azure

PROJECTS

Multimodal AI Search Engine | *PyTorch, FAISS, NumPy, pandas, Streamlit, Python*

Feb. 2025 – Nov. 2025

- Developed a visual search engine by indexing over 10,000 images using CLIP embeddings and FAISS, retrieving semantically relevant images from natural language prompts with an average query time of 2 seconds.
- Built and deployed a Streamlit interface for real-time, scrollable image results, demonstrating practical ML deployment and cross-modal embedding techniques.

JotDown | *Swift, SwiftUI, Git, GitHub, Firebase, ML*

Sep. 2025 – Dec. 2025

- Architected the application's UI with SwiftUI, employing an MVVM design pattern to build a responsive and minimalist interface for frictionless data entry.
- Integrated on-device Core ML models to perform Natural Language Processing, enabling a semantic search feature that recognizes entities like dates and keywords within user queries.
- Designed a data visualization module to generate a knowledge graph from user notes, modeling semantic relationships to uncover latent themes and connections.

EXPERIENCE

Georgia Tech Research Institute (GTRI)

May. 2026 – Present

Software Engineer Intern, CIPHER Lab – Threat Analysis Branch

Atlanta, GA

- Developing ML R&D solutions and graph analytics pipelines in support of cybersecurity and intelligence analysis workflows.
- Applying NLP techniques to analyze and extract insights from large-scale unstructured data in a national security context.
- Contributing to a graph analytics dashboard platform for cybersecurity tooling, working with vertex/edge graph data structures and collaborating iteratively with the research team on ML R&D projects.

Georgia Institute of Technology – VIP Program

Jan. 2026 – Present

Automated Algorithm Design – AlphaEvolve Subteam

Atlanta, GA

- Contribute to a GTRI-sponsored research program developing a multi-objective genetic programming framework that automatically evolves hybrid algorithms from data, outperforming human-designed baselines.
- Work on the AlphaEvolve subteam extending the LLM-Guided Evolution (LLM-GE) framework — cited as prior work by Google DeepMind's AlphaEvolve — which uses LLMs to propose, mutate, and evaluate code in an automated evolutionary loop.
- Run and validate experiments on Georgia Tech's PACE ICE HPC cluster, developing reproducible Python workflows for neural architecture search and performance benchmarking.

Georgia Tech iOS Club

Jan. 2026 – Present

Senior iOS Developer

Atlanta, GA

- Led a team of 8 developers to build iOS apps with Swift, Core ML, and Firebase, delivering 3 club projects used by 150+ members with an average 4.7/5 satisfaction rating.
- Collaborated with designers and developers to architect SwiftUI features using MVVM, reducing onboarding drop-off by 30% and increasing average session duration by 2 minutes.
- Reviewed 40+ pull requests in GitHub, provided technical feedback using code-review guidelines, and mentored 4 junior developers, reducing bug rate by 25% and cutting feature delivery time by 1 week.